

Computational dynamics

Computer Laboratory 2:

Introduction to Elastodynamics in FEniCSx.
Time Discretization & Numerical/Rayleigh Damping

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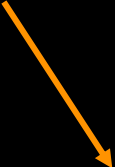
INTRODUCTION

In structure dynamics we often work with systems of form

$$\mathbf{M} \cdot \ddot{\mathbf{d}}(t) + \mathbf{C} \cdot \dot{\mathbf{d}}(t) + \mathbf{K} \cdot \mathbf{d}(t) = \mathbf{F}(t)$$

$$\mathbf{d}(0) = \mathbf{d}_0$$

$$\dot{\mathbf{d}}(0) = \mathbf{v}_0$$



Viscous damping
matrix

- Discrete damping
- Material /visco damping
- Damping due to contact / plasticity
- **A) NUMERICAL DAMPING**
- ...
- **B) RAYLEIGH DAMPING:**

$$\mathbf{C} = a\mathbf{M} + b\mathbf{K}$$



Parameters

Determined from:

- Experience
- Literature / technical guidelines (DIN, ASME, KTA, ...)

[Modal reduction to SDOF: done on blackboard]

$$\text{Damping constant: } \xi_{(l)} = \frac{1}{2} \left[\frac{a}{\omega_{(l)}} + b\omega_{(l)} \right]$$

A) NUMERICAL DAMPING

- Numerical damping refers to the **artificial dissipation of energy** introduced by the Newmark method, especially **at high frequencies**.
- Can be either desired (to suppress spurious oscillations) or unwanted (distorts the physical response).
- Depends on Δt and vanishes for $\Delta t \rightarrow 0$. Note: stable method but big Δt can produce a completely wrong (but stable) solution!
- Whether and how numerical damping appears **depends on the choice of Newmark parameters**.

Newmark update equations:

$$u_{n+1} = u_n + \Delta t \dot{u}_n + \frac{\Delta t^2}{2} [1 - 2\beta] \ddot{u}_n + 2\beta \ddot{u}_{n+1}$$
$$\dot{u}_{n+1} = \dot{u}_n + \Delta t [1 - \gamma] \ddot{u}_n + \gamma \ddot{u}_{n+1}$$

1) **No damping** (second-order accurate, unconditionally stable, energy-conserving):

$$\gamma = \frac{1}{2}; \quad \beta = \frac{1}{4}$$

2) **With damping to suppress high-frequency spurious oscillations** (unconditionally stable):

$$\gamma > \frac{1}{2}; \quad \beta \geq \frac{1}{4} \left[\gamma + \frac{1}{2} \right]^2; \quad \gamma \text{ is fundamental in controlling numerical damping.}$$

Note: low frequencies are barely affected!

A) NUMERICAL DAMPING

```
##### NEWMARK METHOD (time discretisation) #####  
HIGH-FREQUENCY DISSIPATION (NUMERICAL DAMPING) IN NEWMARK  
METHOD:  
gamma = 1  
beta = 0.5
```

Some examples:

γ	β	Behaviour
0.5	0.25	No numerical damping
0.6	0.3025	Mild high-frequency damping
1.0	0.5	Strong high-frequency damping
1.0	1.0	Not necessarily “twice as much damping”; accuracy and dispersion also change

3) Tuned damping with **Generalized- α method**:

$$\gamma = \frac{1}{2} + \alpha; \quad \beta = \frac{1}{4} [1 + \alpha]^2, \quad \alpha \in [0, 0.3]$$

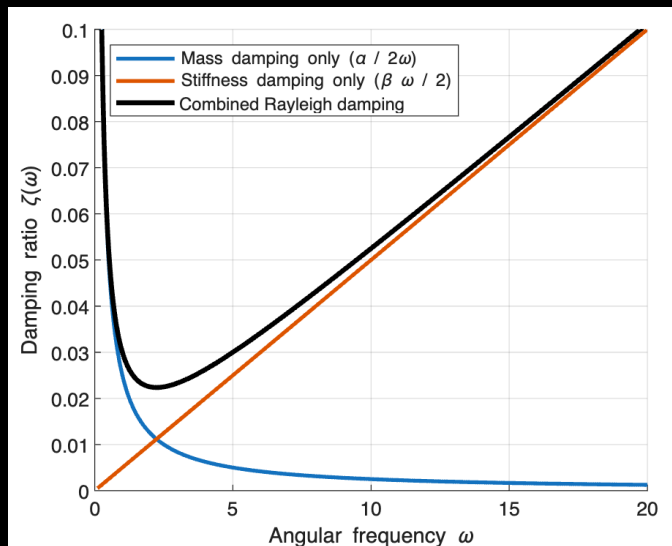
- Controls high-frequency dissipation more systematically;
- Often preferred because it damps only the unwanted high frequencies while preserving the low-frequency physical response more accurately.

B) RAYLEIGH DAMPING

$$\mathbf{M} \cdot \ddot{\mathbf{d}}(t) + \mathbf{C} \cdot \dot{\mathbf{d}}(t) + \mathbf{K} \cdot \mathbf{d}(t) = \mathbf{F}(t)$$

$$\mathbf{C} = \eta_m \mathbf{M} + \eta_k \mathbf{K}$$

$$\text{Damping constant: } \xi_{(l)} = \frac{1}{2} \left[\frac{\eta_m}{\omega_{(l)}} + \eta_k \omega_{(l)} \right]$$



$\eta_m \mathbf{M}$: Damping at low frequencies.

$\eta_k \mathbf{K}$: Damping at high frequencies.

```
##### RAYLEIGH DAMPING #####
# Note: Not necessary for HF dissipation if γ and β are properly chosen.
eta_m = 0 #1e-4
eta_k = 0 #1e-4
```

```
def avg(x_old,x_new,alpha):
    return alpha*x_old + (1-alpha)*x_new
def update_a(u,u_old,v_old,a_old):
    dt_ = dt
    beta_ = beta
    return (u-u_old-dt_*v_old)/beta_/dt_**2 - (1-2*beta_)/2/beta_*a_old
def update_v(a,u_old,v_old,a_old):
    dt_ = dt
    gamma_ = gamma
    return v_old + dt_*((1-gamma_)*a_old + gamma_*a)
a_new = update_a(u,u_old,v_old,a_old)
v_new = update_v(a_new,u_old,v_old,a_old)
```

```
# Reminder
Fv = I + ufl.grad(avg(v_old,v_new,alpha_f))
```

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##### VARIATIONAL PROBLEM – WEAK FORM
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```
Res_u =
    rho * ufl.inner(avg(a_old,a_new,alpha_m),v)) * dx\
+ eta_m * rho * ufl.inner(avg(v_old,v_new,alpha_m),v)\
+ eta_k * (ufl.inner(Piso(Fv, 0), ufl.grad(v)) + ufl.inner(Pvol(Fv, 0),
ufl.grad(v)))\
+ (ufl.inner(Piso(F, 0), ufl.grad(v)) + ufl.inner(Pvol(F, 0), ufl.grad(v))
- ufl.inner(traction,v) * dss(1))
```

C) COMPARISON

NUMERICAL DAMPING:

- Acts on the (time) discrete system
- No physical meaning
- Dependence on Δt ; vanishes for $\Delta t \rightarrow 0$
- Good to damp high frequencies

RAYLEIGH DAMPING:

- Acts on the (time) continuous system
- Approximate physical dissipation
- No dependence on Δt ; remains for $\Delta t \rightarrow 0$
- Good to damp across wider frequency range

TASKS

1. Explore high-frequency spurious oscillations with

Wait until
> 7 cycles.

No damping (second-order accurate, unconditionally stable, energy-conserving): $\gamma = \frac{1}{2}$; $\beta = \frac{1}{4}$.

2. Suppress high-frequency oscillations with numerical damping

$$\gamma > \frac{1}{2}; \beta \geq \frac{1}{4} \left[\gamma + \frac{1}{2} \right]^2.$$

Compare 2 cases(*).

- Does it eliminate spurious oscillations?
- Does it introduce damping in the oscillation?

3. Add Rayleigh damping.

Choose parameters η_m and η_k . Compare 2 cases(*).

$$\mathbf{C} = \eta_m \mathbf{M} + \eta_k \mathbf{K}$$

Call the instructor for doubts and after successful completion of each task and report it. Enjoy!

TASKS

(*) Compare 2 cases:

FOR 2) Set γ and β according to your *Matrikelnummer*.

Take the last 2 digits: d_1, d_2 . Compute:

$$\gamma = \frac{1}{2} + 0.05 d_1 \quad \text{and} \quad \beta = \frac{1}{4} \left[\gamma + \frac{1}{2} \right]^2 + 0.01 d_2$$

FOR 3) Set η_m and η_k according to your *Matrikelnummer*.

Take the last 3 digits: d_1, d_2, d_3 . Compute:

$$\eta_m = [d_1 + d_2] 10^{-4} \quad \text{and} \quad \eta_k = d_3 10^{-4}$$